

A Technical Introduction to Vision transformer

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Section 1

A vision transformer (ViT) is a transformer designed for computer vision. A ViT decomposes an input image into a series of patches (rather than text into tokens), serializes each patch into a vector, and maps it to a smaller dimension with a single matrix multiplication. These vector embeddings are then processed by a transformer encoder as if they were token embeddings. ViTs were designed as alternatives to convolutional neural networks (CNNs) in computer vision applications. They have different inductive biases, training stability, and data efficiency. Compared to CNNs, ViTs are less data efficient, but have higher capacity. Some of the largest modern computer vision models are ViTs, such as one with 22B parameters. Subsequent to its publication, many variants were proposed, with hybrid architectures with both features of ViTs and CNNs. ViTs have found application in image recognition, image segmentation, weather prediction, and autonomous driving.

Section 2

Others Other examples include the visual transformer, CoAtNet, CvT, the data-efficient ViT (DeiT), etc. In the Transformer in Transformer architecture, each layer applies a vision Transformer layer on each image patch embedding, add back the resulting tokens to the embedding, then applies another vision Transformer layer.

Section 3

Comparison with CNNs Typically, ViT uses patch sizes larger than standard CNN kernels (3x3 to 7x7). ViT is more sensitive to the choice of the optimizer, hyperparameters, and network depth. Preprocessing with a layer of smaller-size, overlapping (stride < size) convolutional filters helps with performance and stability. This different behavior seems to derive from the different inductive biases they possess:

Section 4

The original ViT was an encoder-only Transformer supervise-trained to predict the image label from the patches of the image. As in the case of BERT, it uses a special token <CLS> in the input side, and the corresponding output vector is used as the only input of the final output MLP head. The special token is an architectural hack to allow the model to compress all information

relevant for predicting the image label into one vector. Transformers found their initial applications in natural language processing tasks, as demonstrated by language models such as BERT and GPT-3. By contrast the typical image processing system uses a convolutional neural network (CNN). Well-known projects include Xception, ResNet, EfficientNet, DenseNet, and Inception. Transformers measure the relationships between pairs of input tokens (words in the case of text strings), termed attention. The cost is quadratic in the number of tokens. For images, the basic unit of analysis is the pixel. However, computing relationships for every pixel pair in a typical image is prohibitive in terms of memory and computation. Instead, ViT computes relationships among pixels in various small sections of the image (e.g., 16x16 pixels), at a drastically reduced cost. The sections (with positional embeddings) are placed in a sequence. The embeddings are learnable vectors. Each section is arranged into a linear sequence and multiplied by the embedding matrix. The result, with the position embedding is fed to the transformer.

Section 5

CNN applies the same set of filters for processing the entire image. This allows them to be more data efficient and less sensitive to local perturbations. ViT applies self-attention, allowing them to easily capture long-range relationships between patches. They also require more data to train, but they can ingest more training data compared to CNN, which might not improve after training on a large enough training dataset. ViT also appears more robust to input image distortions such as adversarial patches or permutations.

Section 6

LayerNorm immediately after each attention and feedforward layer ("res-post-norm"); scaled cosine attention to replace the original dot product attention; log-spaced continuous relative position bias, which allows transfer learning across different window resolutions.

Section 7

Further reading Zhang, Aston; Lipton, Zachary; Li, Mu; Smola, Alexander J. (2024). "11.8. Transformers for Vision". Dive into deep learning. Cambridge New York Port Melbourne New Delhi Singapore: Cambridge University Press. ISBN 978-1-009-38943-3. Steiner, Andreas; Kolesnikov, Alexander; Zhai, Xiaohua; Wightman, Ross; Uszkoreit, Jakob; Beyer, Lucas (June 18, 2021). "How to train your ViT? Data, Augmentation, and Regularization in Vision Transformers". arXiv:2106.10270 [cs.CV].

Section 8

History Transformers were introduced in Attention Is All You Need (2017), and have found widespread use in

natural language processing. A 2019 paper applied ideas from the Transformer to computer vision. Specifically, they started with a ResNet, a standard convolutional neural network used for computer vision, and replaced all convolutional kernels by the self-attention mechanism found in a Transformer. It resulted in superior performance. However, it is not a Vision Transformer. In 2020, an encoder-only Transformer was adapted for computer vision, yielding the ViT, which reached state of the art in image classification, overcoming the previous dominance of CNN. The masked autoencoder (2022) extended ViT to work with unsupervised training. The vision transformer and the masked autoencoder, in turn, stimulated new developments in convolutional neural networks. Subsequently, there was cross-fertilization between the previous CNN approach and the ViT approach. In 2021, some important variants of the Vision Transformers were proposed. These variants are mainly intended to be more efficient, more accurate or better suited to a specific domain. Two studies improved efficiency and robustness of ViT by adding a CNN as a preprocessor. The Swin Transformer achieved state-of-the-art results on some object detection datasets such as COCO, by using convolution-like sliding windows of attention mechanism, and the pyramid process in classical computer vision.

Section 9

The basic architecture, used by the original 2020 paper, is as follows. In summary, it is a BERT-like encoder-only Transformer. The input image is of type $\mathbb{R}^{H \times W \times C}$, where H, W, C are height, width, channel (RGB). It is then split into square-shaped patches of type $\mathbb{R}^{P \times P \times C}$. For each patch, the patch is pushed through a linear operator, to obtain a vector ("patch embedding"). The position of the patch is also transformed into a vector by "position encoding" (the paper tried no embedding, 1D embedding, 2D embedding, and relative embedding: 1D was adopted). The two vectors are added, then pushed through several Transformer encoders. The attention mechanism in a ViT repeatedly transforms representation vectors of image patches, incorporating more and more semantic relations between image patches in an image. This is analogous to how in natural language processing, as representation vectors flow through a transformer, they incorporate more and more semantic relations between words, from syntax to semantics. The above architecture turns an image into a sequence of vector representations. To use these for downstream applications, an additional head needs to be trained to interpret them. For example, to use it for classification, one can add a shallow MLP on top of it that outputs a probability distribution over classes. The original paper uses a linear-GeLU-linear-softmax network.